



The conditioning of surprisal and many-to-one mappings in speech variation: Articulatory efficiency and perceptual recoverability

Po-Hsuan Huang¹ & Canaan Breiss^{1,2}

¹ University of Southern California

² University of Chicago



USC

LabPhon 20, Montréal, June 26–28 2026



Speech variation under information-theoretic constraints

- Different information-theoretic (Shannon 1948) correlates are shown to predict different types of variation:
- More informative/unpredictable items maintain higher distinctiveness/are more resilient to variation.
- For example:
 - Frequencies correlate w/ word durations (Baker & Bradlow 2009) & vowel merger rates (Kang, Yoon & Han 2015).
 - Functional load (# of minimal pairs) correlates w/ merger rates & vowel lengths (Wedel et al. 2013).